Prabhavit Gupta



Principal AI & Analytics Engineer: Reporting Platforms & AI-Driven Engineering

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (CET Time Zone / EU Permanent Resident / Available Immediately)

Summary

Hands-on Principal AI and Analytics Platform Engineer with 3 years of commercial software engineering experience architecting high-performance reporting systems, analytical data models, and modern multi-tenant SaaS data platforms. Pioneer in leveraging AI coding tools (Claude Code, Cursor) to automate and accelerate end-to-end software delivery from greenfield design to production release. Proven track record designing self-service report builders, optimizing high-volume PostgreSQL analytical queries, enforcing strict tenant isolation, and implementing automated data reconciliation test suites. Based in Venice, Italy (CET time zone) with an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required); available immediately for full-time remote roles.

Technical Expertise

- **Analytics & Reporting Architecture:** Self-service reporting engines, dimensional data modeling, aggregation pipelines, metrics catalogs, SQL query optimization, automated data reconciliation test suites.
- **AI-Accelerated Software Development:** Claude Code autonomous workflows, Cursor AI pair programming, Model Context Protocol (MCP) tool integration, prompt regression testing.
- **Backend & Cloud Systems:** Python (FastAPI, asyncio, Pydantic), TypeScript/Node.js, PostgreSQL (partitioning, indexing, tenant schema isolation), AWS cloud services, Redis in-memory caching.
- **Multi-Tenant Security & Reliability:** Strict tenant isolation, role-based access control (RBAC), audit logging, zero-downtime database migrations, 99.8% production SLA maintenance.
- **Testing & Observability:** End-to-end regression testing with pytest and Playwright, Prometheus telemetry, structured logging, CI/CD with GitHub Actions.
- **Languages & Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Conversational). EU Long-Term Residence Permit (unrestricted EU work authorization, no sponsorship required). Available immediately.

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Analytics & AI Systems Engineer *London, UK (remote)*

- Designed and delivered the core analytical data pipeline and customer reporting infrastructure serving 4,500+ enterprise users with sub-second report generation.
- Accelerated full-stack platform development using Claude Code and Cursor, authoring automated test suites, schema migrations, and REST APIs at 3x standard velocity.
- Architected multi-tenant PostgreSQL data models with strict tenant isolation, decreasing analytical query p95 latency by 42% through strategic composite indexing.
- Built automated data reconciliation engines comparing transactional source tables against materialized reporting aggregates to guarantee 100% metric fidelity.
- Deployed containerized microservices and automated CI/CD workflows on cloud infrastructure, sustaining 99.8% availability across continuous release cycles.

Urlink **Apr 2025 - Jul 2025**
Analytics Platform Developer (Freelance) *Barcelona, Spain (remote)*

- Implemented automated data ingestion and reporting transformations extracting structured operational metrics into centralized analytical stores.
- Built automated regression and data integrity test suites with pytest and Playwright to validate calculated KPIs against benchmark operational datasets.
- Developed custom export utilities allowing enterprise stakeholders to schedule and download self-service CSV and PDF reports.

Independent Technical Consultant **Aug 2023 - Feb 2024**
Data Systems & Software Engineer *Venice, Italy*

- Designed and deployed custom reporting dashboards and ETL microservices for European SMB clients, automating manual spreadsheet reporting workflows.

- Built optimized SQL queries and database migration pipelines in Python for multi-tenant customer management applications.
- Authored technical architecture runbooks and data dictionary documentation for client engineering leadership.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous training in formal logic, statistical inference, quantitative economics, and complex systems modeling.